Physics Worksheet

Waves

Name: ____________________ **Student #:** _______________ **Student Email #:** ________

1. Rank the following mediums from fasted to slowest wave speed and explain why: Ice, vapour, water, vaccuum of space

2. Imagine someone is strumming a guitar infront of you

   (a) Which strings on the guitar make a higher pitched noise and why?

   (b) If all the strings are strum at the same time, which string will you hear the noise from first?

3. Show that the work done by a spherical wave over the course of its period T in a fluid is: $W = \omega^3 A^2 r^2 \sqrt{B\rho}$

4. In movies, when a glass shatters due to someone singing a very high note, what are the theoretical physics behind what is happening?

5. For a standing wave on a string, at which harmonic will the velocity be such that it travels across the full lenght of string L in one period T?